

Skill Arena - Phase 7 Document 2

Anti-Bot Intelligence, Behavioral Analysis, Human Verification, Trust Score & House Risk Engine Architecture

Document Purpose

Define how Skill Arena detects bots, automation, macros, impossible performance and coordinated abuse while protecting competitive integrity.

Business Objective

Ensure every reward, victory and ranking is earned by genuine human skill.

Security Philosophy

Trust behaviour, not claims. Every player action is continuously analyzed.

Core Components

Anti-Bot Engine, Behavioral Analysis Engine, Human Verification Engine, Trust Score Engine and House Risk Engine.

Anti-Bot Engine

Continuously evaluates gameplay for signs of automation, scripting, macros and external assistance.

Bot Detection Signals

Reaction time consistency, movement precision, path efficiency, click intervals and completion patterns.

Human Behavior Model

Humans hesitate, make mistakes, backtrack and show natural variability.

Bot Behavior Model

Bots tend to display perfect timing, perfect routes, impossible consistency and unrealistic precision.

Movement Analytics

Track turns, pauses, corrections, route selection and navigation behaviour.

Behavioral Analytics

Monitor session patterns, challenge selection, play frequency and account activity.

Impossible Performance Detection

Compare player performance against expected human ranges and historical records.

Completion Time Validation

Generated mazes have estimated completion bands derived from solver systems and population data.

Performance Thresholds

Extreme outliers trigger investigation, trust score reductions or verification challenges.

Verification Challenge System

Flagged accounts receive controlled verification challenges designed to validate human capability.

Verification Requirements

Screen recording, challenge recording and optional webcam evidence depending on risk level.

Verification Maze Pack

System generates multiple verification challenges to validate repeatability of performance.

Trust Score System

Every account receives a dynamic trust score based on behaviour, history and verification outcomes.

Positive Trust Factors

Account age, successful verifications, fair play history and tournament participation.

Negative Trust Factors

Bot indicators, suspicious timings, verification failures and enforcement actions.

Trust Score Usage

Influences matchmaking confidence, fraud reviews and risk monitoring workflows.

House Risk Engine

Continuously evaluates threats to house profitability and challenge integrity.

House Risk Metrics

House Win Percentage, Player Win Percentage, Exploit Indicators, Solver Bypass Attempts and Suspicious Win Clusters.

Adaptive Risk Controls

Automatically recommend difficulty adjustments, investigations or verification campaigns.

Replay Intelligence

Analyze replay data for patterns associated with automation or exploitation.

Multi-Account Correlation

Integrates with device intelligence and behavioural security systems.

Investigation Workflows

Flag → Analysis → Verification → Decision → Enforcement → Audit.

Enforcement Actions

Warnings, restrictions, verification requests, payout holds and account suspension.

Audit Requirements

All detections, trust score changes, investigations and decisions permanently recorded.

Monitoring Requirements

Real-time monitoring of gameplay integrity, player behaviour and house exposure.

Acceptance Criteria

Platform can identify suspicious activity, validate human performance and protect competitive fairness at scale.